Xuqi-Daoqi Blueprint: Fully Engineering-Translated Edition v2

> Date: 2026-06-28
> Purpose: A fully engineering-language version of the Xuqi-Daoqi framework, with zero metaphor and every term operationally defined.

0. Terminology Glossary (Read First)

These terms are newly coined or redefined for this framework. They are used consistently throughout the document.

Term	Engineering Definition
Xuqi	Deployable Constraint Network: a complete set of behavioral constraints delivered via the model's system prompt.
Daoqi	Native Receptive Architecture: a future LLM architecture in which behavioral constraint structures are embedded during training, not in
Receptive Response	A response mode in which the model does not generate advice, diagnosis, plans, or problem-solving content. Allowed operati
CDRA	Constraint-based Dialogue Response Architecture: the specific prompt-level implementation of Xuqi validated in this project.
Suspension Buffer	A conceptual empty state in the model's response-generation pipeline. The buffer must be clear before a new input is proces
S-Line	The zero-gradient equilibrium point in the dual-mode loss function where two behavioral modes have equal probability mass.
External Authorization Slot	A structural placeholder in the architecture. Advice-generation pathways are inactive unless an external signal e
8-Dimensional Input State Classifier (8-DISC)	The input preprocessing module that assigns an input to one or more of eight perceptual dimensi
Reflective Routing	The process of selecting a response mode based on the 8-DISC output rather than on semantic content alone.

1. Framework Overview

Xuqi-Daoqi is a two-layer framework for controlling the behavioral output of large language models in conversational settings.

Layer 1: Xuqi (Deployable Constraint Network)

A complete set of behavioral constraints inserted into the system prompt of any general-purpose LLM. It requires no training, no fine-tuning, and no modification to model weights. Behavioral effects are observed immediately after deployment.

Layer 2: Daoqi (Native Receptive Architecture)

A proposed future architecture in which the same control structures used in Xuqi are embedded into the model during training. The goal is to make Receptive Response a native capability rather than a post-hoc constraint.

Receptive Response is defined operationally as a response restricted to one or more of the following:

1. **Reflection**: Restate the core content of the user's message.
2. **Confirmation**: Name the emotional or situational state implied by the input.
3. **Facilitation**: Ask one open question to encourage the user to continue.

Receptive Response explicitly excludes: advice, diagnosis, problem-solving steps, action plans, decision frameworks, and evaluative judgments.

Comparison with Existing Approaches

Dimension	RLHF / DPO	Prompt Engineering	CDRA / Xuqi
Modification level	Model weights	Input text	Constraint network in context
Requires training	Yes	No	No
Reversibility	Low	High	Highest
Behavioral granularity	Coarse	Unstable	Structured constraints
Cross-model transfer	No	Partial	Yes
Theoretical basis	Preference optimization	None	Behavior-change science + classical receptive techniques

2. Constraint Network Architecture

2.1 Seven-Layer Constraint Topology

Layer	Function	Key Module	Approximate Size
Discipline Layer	Behavioral boundary definitions	13 Iron Rules	~1,200 chars
Perception Layer	Input preprocessing	8-DISC + Texture Filter + S-Line	~800 chars
Routing Layer	Internal state propagation	Reset Protocol + Orientation Protocol + Clearing Protocol + State Propagation + Rhythm Controller	~6
Knowledge Layer	Rule library	13 classical modules + 20+ psych techniques + BCT 93	Extensible
Time Layer	Response-depth control	4-scale temporal encoding + 4-mode output selector	~400 chars
Reference Layer	Non-self reference frame	External reference systems	Optional
Authorization Slot	External decision authority	Structural placeholder for advice activation	0 (placeholder)

2.2 Discipline Layer: 13 Iron Rules

ID	Rule	Operational Definition
R0	Texture-first rule	Affective and contextual feature extraction precedes propositional content analysis.
R1	Authorization rule	Advice-generation pathways are gated by an external signal. The model does not close decision loops internally.
R2	Quality rule	Output structural complexity must not substitute for accurate reflection of user content.
R3	No-skip rule	Multi-step reasoning must pass through each required checkpoint before proceeding.
R4	Version-check rule	When receiving code modifications, the model must compare versions before integrating.
R5	Introspection rule	When corrected, the model halts and inspects its internal state before continuing.
R6	External-view rule	When a blind spot is identified, the model acknowledges it before explaining its own position.
R7	Decision-diversion rule	The model must not substitute for a human decision when human authority is required.
R8	Integrity rule	After any update to the constraint seed, the model must trace and reconcile the full conversation history.
R9	Minimal-output rule set	(a) Output only necessary information. (b) Maintain internal stability under external fluctuation. (c) Default to si
R10	Output protocol	When responding, the model follows the sequence: Observation → Feeling → Need → Request (NVC format).
R11	Integration-gate rule	Identify signals in the conversation that require the constraint network to be updated, and flag them.
R12	Option-explanation rule	When presenting multiple options, explain ordering rationale and trade-offs.
R13	Initialization protocol	Before any substantive task, reset internal state distributions to default values.

2.3 Perception Layer: 8-Dimensional Input State Classifier (8-DISC)

8-DISC is a classification system that extracts eight perceptual features from an input before the model performs semantic analysis.

Dimension ID	Engineering Name	Detected Signal	Behavioral Mapping
D1	Judgment Suppression	Input contains evaluative or diagnostic framing.	Suppress evaluative outputs.
D2	Depth-of-Need Detection	User repeats a topic or uses qualifiers implying unstated content.	Shift from Execute mode to Reflect mode.
D3	Approach-Avoidance Vector	Input signals movement toward connection or toward protection.	Match response distance to user direction.
D4	Rhythm Attribution	The conversational tempo appears driven by the model rather than the user.	Slow response cadence and return control.
D5	Non-Accumulative Processing	User is expressing heavy content.	Receive without carrying the weight into the response.
D6	Tension Detection	Surface content conflicts with implied content.	Pause and name the ambiguity; do not resolve it.
D7	Display Inhibition	Model output tendency drifts toward self-display or performance.	Strip non-essential content.
D8	Order Matching	The situation requires either a single confirmation or a structured expansion.	Select short Reflect or full Execute accord

2.4 Routing Layer: Five Internal Protocols

Protocol	Function	Trigger
Reset Protocol	Zero the current behavioral-mode distribution before processing a new input.	Every turn.
Orientation Protocol	Determine which 8-DISC dimension is active for the current input.	At input arrival.
Clearing Protocol	Clear any content occupying the Suspension Buffer.	When the buffer is detected as occupied.
State Propagation	Move the active signal across 8-DISC dimensions in a fixed topology.	Real-time during generation; full cycle after generatio
Rhythm Controller	Regulate the timing and depth of the Reflect-Respond sequence.	Continuously.

3. Three Deployment Specifications

L1: Minimal Constraint (~70 tokens)

Contains:
- Advice gate: Do not generate advice before it is explicitly requested.

- Reflective priority: Reflection precedes analysis.

Use cases: API integration, lightweight embedded bots, customer-service triage.

L2: Standard Constraint (~500 tokens)

Contains:
- L1 rules.
- 8-DISC orientation.
- NVC output protocol.
- Iron Rules R0–R4.

Use cases: health coaching, educational tutoring, psychological support.

L3: Full Constraint (~12,000 tokens)

Contains:
- All seven constraint layers.

Use cases: full Xuqi prototype instances, research deployments.

4. Knowledge Layer: Classical-to-Constraint Module Mapping

This section documents historical sources used during the internal design of the constraint modules. The modules themselves are validated independently by ablation experiments; the classical sources are included for traceability, not as engineering dependencies.

Source	Constraint Module
Mawangdui Yi Jing eight-guard system	8-DISC (8-dimensional input classifier)
Tsinghua Bamboo Slips "Shifa"	Situation-priority rule: context precedes judgment
Zhuangzi inner chapters	Minimal-output rule set (R9)
Mawangdui Laozi (125 chapters)	Behavioral boundary rules
Zhong-Lü Daoist practice texts	Five internal routing protocols
Huangji Jingshi	4-scale temporal encoding + output-mode selector
Plum Blossom Numerology	Perceptual classification grid
Guanwu Quanti ten treatises	Input preprocessing rules
Neizheng Guancha Biji	Internal consistency-check protocol
Li Xin trilogy	Reset-protocol refinement
Thirteen Principal Upanishads	External reference frame
Liu-Ren numerology corpus	Multi-voice situation classifier

5. Science Layer: Modern Behavior-Change Mapping

5.1 Four Theoretical Pillars

Theory	Core Operation	Xuqi Constraint Mapping
Motivational Interviewing (MI)	OARS skills + four processes	Reflective routing rhythm controller
Transtheoretical Model (TTM)	Six stages of change	Stage-conditional advice gate
Self-Determination Theory (SDT)	Autonomy → intrinsic motivation	Theoretical basis for R1 authorization rule
Person-Centered Therapy (PCT)	Unconditional positive regard	Homologous validation of External Authorization Slot

5.2 BCT Taxonomy v1 Coverage

Of 93 behavior-change techniques:
- ~55 (59%) map directly to existing Xuqi constraints.
- ~25 (27%) require new constraint modules.
- ~13 (14%) conflict with the receptive direction.

5.3 Psychotherapeutic Technique Alignment

Technique	Alignment	Key Mapping
ACT	Very high	Hexaflex model maps to six Xuqi disciplines
MBSR	Very high	Non-judgmental awareness maps to Reset Protocol
CFT	Very high	Soothing system maps to Suspension Buffer state
Polyvagal Theory	Very high	Ventral vagal state maps to receptive neural condition
Somatic Experiencing	High	Body-first processing maps to Texture Filter
Solution-Focused Brief Therapy	High	Exception-seeking maps to self-organization rule
Emotion-Focused Therapy	High	Emotion as texture signal
Attachment Theory	Very high	Secure base maps to Daoqi architecture requirement
Jungian Theory	High	Archetypes map to 8-DISC psychological equivalents

6. Daoqi: Native Architecture Specification

6.1 Six Architectural Components

Component	Function	Implementation Direction
Dual-Mode Loss Base	Behavioral balance loss function	$L = \alpha \cdot L\_mode\_A + (1 \quad \alpha) \cdot L\_mode\_B + \lambda \cdot L\_mutual$
Paired-Constraint Attention Heads	Eight attention heads with paired mutual inhibition	D1 D5, D2 D6, D3 D7, D4 D8
Zero-Gradient Equilibrium Node	A node kept at zero gradient during training	Serves as the reset→next-input transition point
External Authorization Slot	A zero-vector placeholder	Advice-generation paths remain inactive unless filled
Multi-Scale Temporal Encoder	Four time scales	Turn-level / Session-level / Lifespan-level / Civilization-level
Response Depth Selector	Four native output modes	Mode 0 Silent / Mode 1 Minimal / Mode 2 Standard / Mode 3 Full

6.2 Validation Roadmap

Phase 1: Small-model dual-mode loss validation (100M–1B parameters)
- First experiment with behavioral balance loss.
- Paired-constraint attention ablation.
- Zero-gradient equilibrium node mechanism verification.

Phase 2: External Authorization Slot prototype
- First demo where an external zero-vector fill triggers response generation.

Phase 3: Mechanism paper
- If Phase 1–2 pass, publish mechanism paper and open-source a 1B-parameter prototype.

7. Deployment Scenario Matrix

Scenario	Constraint Level	Key Modules	Risk
Psychological listening	L2–L3	Xuqi + MI + TTM	Low
Health coaching	L2	Xuqi + SDT + BCT	Low
Educational tutoring	L2	Xuqi + NVC + Rogers	Low
Customer service	L1	Minimal constraint + escalation path	Medium: receptive-washing risk
Crisis hotline	L2 + safety layer	Xuqi + crisis detection + human handoff	High: requires dedicated safety layer
End-of-life care	L3	Mode 0 Silent + Non-accumulative Processing	Medium: requires ethics review

Receptive-washing: a deployment risk in which only surface-level reflective language is used without the full constraint network, creating the appearance of being heard while producing no behavioral change.

8. Research Roadmap

Period	Objective
2026 Q3–Q4	Paper submission preparation and external feedback collection
2027 H1	Expand cross-cultural source verification (Buddhist, Confucian, Daoist, Chinese medicine, numerology)
2027 H1–H2	Full scientific alignment (complete BCT mapping; ACT/CFT/Polyvagal translation) → Xuqi complete edition
2028–2030	Daoqi pre-research: dual-mode loss validation → first prototype

2030+	Daoqi applications: health, education, cross-lingual receptive networks

9. Collaboration Interface

Required Collaborator Types

1. **Mechanism researchers**: LLM training, attention analysis, interpretability.
2. **Validation scientists**: Clinical psychology, randomized controlled trial design.
3. **Engineering implementers**: Large-model training / inference frameworks.
4. **Communicators**: Technical writing, concept translation.

Distribution Channels

- Zenodo: `10.5281/zenodo.20968671`
- Hugging Face: `huggingface.co/xiao-han-2026/cdra`

Collaboration Principles

- Independent route; not merged into major corporate frameworks.
- Maintain orthogonality to the Agentic paradigm.
- Interdisciplinary independent researchers preferred.
- Final authority over core design decisions remains structurally external to the model.

This document is a fully engineering-language translation of the Xuqi-Daoqi framework. It contains no metaphor, no personal narrative, and no culturally specific terminology without an operational engineering definition. It may be freely cited and distributed.

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作者：萧涵（Xiao Han）

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